

LC-3 Instruction Encodings

Arith./Logic		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	ADD†	0	0	0	1	DR		SR1		0			0	0	SR2									
	ADD†	0	0	0	1	DR		SR1		1	imm5													
	AND†	0	1	0	1	DR		SR1		0			0	0	SR2									
	AND†	0	1	0	1	DR		SR1		1	imm5													
	NOT†	1	0	0	1	DR		SR		1			1	1	1	1	1							
Load/Store		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	LD†	0	0	1	0	DR		PCoffset9																
	ST	0	0	1	1	SR		PCoffset9																
	LDI†	1	0	1	0	DR		PCoffset9																
	STI	1	0	1	1	SR		PCoffset9																
	LDR†	0	1	1	0	DR		BaseR		offset6														
	STR	0	1	1	1	SR		BaseR		offset6														
Branch		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	JMP	1	1	0	0	0		0	0	BaseR		0						0	0	0	0			
	BR	0				0	0	0	cn	cz	cp	PCoffset9												
	JSRR	0	1	0	0	0		0	0	BaseR		0						0	0	0	0	0		
	JSR	0				1	0	0	1	PCoffset11														
Misc.		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	LEA	1	1	1	0	DR		PCoffset9																
	TRAP	1	1	1	1	0			0	0	0	trapvect8												
	RTI	1	0	0	0	0												0	0	0	0	0	0	
	illegal	1	1	0	1	x												x	x	x	x	x	x	x

[†] Updates condition code (CC)

LC-3 MMIO Registers

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFE00	KBSR	Rdy	(undefined)														
0xFE02	KBDR	0 0 0 0 0 0 0 0								ASCII code							
0xFE04	DSR	Rdy	(undefined)														
0xFE06	DDR	0 0 0 0 0 0 0 0								ASCII code							
0xFFFC	PSR	Priv	0 0 0 0				PRIO			0 0 0 0 0				N	Z	P	

LC-3 Trap Vectors

- 0x20 → GETC
- 0x21 → PUTC
- 0x22 → PUTS
- 0x25 → HALT

LC-3 Memory Map

0x0000	Trap vector table
0x00FF	
0x0100	Interrupt vector table
0x01FF	
0x0200	OS memory ↑ Supervisor Stack
0x2FFF	
0x3000	Code User memory ↑ User stack
0xFDFF	MMIO
0xFE00	
0xFFFF	

LC-3 Datapath Control Signals

(Use these for filling out LC-3 FSM state transition diagrams)

Signals to assert:

- LD.MAR
- LD.MDR
- LD.IR
- LD.REG
- LD.CC
- GatePC
- GateMDR
- GateALU
- GateMARMUX
- GateSP
- GatePSR
- MIO.EN
- ACK

Signals with named values:

- PCMUX: PC+1, BUS, ADDER
- SR1MUX: IR[11:9], IR[8:6]
- DRMUX: IR[11:9], 7
- ALUK: ADD, AND, NOTA, PASSA
- ADDR1MUX: PC, BaseR
- ADDR2MUX: ZERO, offset6, PCoffset9, PCoffset11
- MARMUX: IR[7:0], ADDER
- SPMUX: Saved.USB, Saved.SSP
- MIO.RW: R, W

LC-3 Microcode Control Signal Bit Assignments

(Use these for filling out bits in microcode tables)

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| <ul style="list-style-type: none">• PCMUX:<ul style="list-style-type: none">• 00₂ → PC+1• 01₂ → BUS• 10₂ → ADDER• SR1MUX:<ul style="list-style-type: none">• 00₂ → IR[11:9]• 01₂ → IR[8:6]• 10₂ → 6• DRMUX:<ul style="list-style-type: none">• 00₂ → IR[11:9]• 01₂ → 7• 10₂ → 6• ALUK:<ul style="list-style-type: none">• 00₂ → ADD• 01₂ → AND• 10₂ → NOTA• 11₂ → PASSA | <ul style="list-style-type: none">• SPMUX:<ul style="list-style-type: none">• 0 → Saved.USB• 1 → Saved.SSP• ADDR1MUX:<ul style="list-style-type: none">• 0 → PC• 1 → BaseR• ADDR2MUX:<ul style="list-style-type: none">• 00₂ → ZERO• 01₂ → offset6• 10₂ → PCoffset9• 11₂ → PCoffset11• MARMUX:<ul style="list-style-type: none">• 0 → IR[7:0]• 1 → ADDER• MIO.RW:<ul style="list-style-type: none">• 0 → R• 1 → W |
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